

MULTI-STEP AI WORKFLOW

ASSESSMENT SUBMISSION

Scenario: AI Study Planner — From Student Requirements to a Refined Personalized Plan

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Assessment Format	Chained multi-prompt AI workflow

Assessment Overview

This assignment demonstrates a three-stage AI workflow in which each prompt depends on the output of the previous stage. The workflow first extracts study requirements, then turns them into a weekly timetable, and finally reviews and refines the timetable using explicit quality criteria.

Workflow: Requirements and planning → Timetable generation → Critique and refinement

Learning objective

To demonstrate how a sequence of connected prompts can transform a broad student need into a practical, checked, and improved study plan. The final schedule is treated as a draft that must be reviewed rather than accepted without evaluation.

1. Assessment Understanding & Workflow Design

The task is designed as a chained workflow rather than three unrelated requests. Prompt 1 identifies constraints and produces a planning blueprint. Prompt 2 uses that blueprint to create the first timetable. Prompt 3 evaluates the timetable against a checklist and produces a corrected version.

Selected scenario

AI Study Planner: create a balanced seven-day study plan for a CSE-AIML student preparing for technical interviews and making progress on academic and project work.

Stage	Purpose	Input	Expected output
1 — Planning	Extract goals, priorities, available time, and scheduling rules.	Student requirements	Structured weekly blueprint
2 — Generation	Convert the blueprint into a usable day-by-day timetable.	Stage 1 blueprint and rules	First-draft weekly schedule
3 — Critique & refinement	Check the draft for balance, time limits, revision, and sequencing.	Stage 2 timetable and review checklist	Critique and final corrected schedule

Student requirements used in this workflow

- Planning horizon: seven days.
- Available focused study time: four hours each weekday and six hours each weekend day.
- Main areas: Java programming, data structures and algorithms (DSA), SQL/DBMS, AI/ML fundamentals, and capstone/project work.
- Preference: short focused sessions with breaks and a small revision period every day.
- Additional goal: use weekend sessions for deeper practice, weak-area correction, and weekly planning.

Why a chained workflow is appropriate

A timetable can become unrealistic if it is generated before the student's constraints are identified. The planning stage makes priorities explicit; the generation stage operationalizes them; and the review stage checks the result for missed requirements and corrects weaknesses. This makes the dependency between stages visible and auditable.

2. Exact Prompt Chain Transcript

The prompts below are designed to be run sequentially in one AI chat. The output from each stage should be retained and used as the input for the next stage.

Prompt 1 — Requirements extraction and planning

Act as an academic planning assistant. I am THUMMAPUDI MANIDEEP, a CSE-AIML student preparing for technical interviews while managing academic learning and capstone/project work. Plan for the next seven days. I can study for four focused hours from Monday to Friday and six focused hours on Saturday and Sunday. My main priorities are Java programming, data structures and algorithms (DSA), SQL/DBMS, AI/ML fundamentals, and project work. I prefer focused sessions of about 50 minutes with short breaks, plus daily active-recall revision. I want the weekend to include deeper practice and review of weak areas.

Task: Extract the requirements, rank the study areas, propose a reasonable weekly time allocation, and create a structured planning blueprint. Do not create the detailed day-by-day timetable yet. Present a table followed by 5–6 rules that can be directly reused in the next prompt.

Prompt 2 — Timetable generation

Using only the planning blueprint and rules produced in Step 1, create the first draft of a seven-day study timetable. Preserve the priority order and the time allocation as closely as possible. Respect four focused hours on each weekday and six focused hours on each weekend day. Include breaks between sessions, daily revision, all five study areas, and manageable project-work blocks. Present a day-by-day table that identifies the focus sessions, and briefly explain how the schedule follows the Step 1 rules. If a detail is not available in the blueprint, state a reasonable assumption instead of silently changing the requirements.

Prompt 3 — Critique gate and refinement

Before finalizing the timetable, act as a strict academic reviewer. Evaluate the Step 2 draft against these criteria: (1) weekday and weekend focused-hour limits are respected; (2) all five priority areas receive meaningful time; (3) each day includes revision; (4) sessions are reasonably short and breaks are included; (5) project work is distributed realistically; and (6) related concept learning is followed by practice where possible.

First, list any failures or weak points and explain the correction for each. Then provide the complete corrected seven-day timetable. Preserve useful parts of the draft, but fix any conflicts with the original requirements. Do not claim that a criterion is satisfied unless the schedule shows it.

3. Intermediate AI Responses

Step 1 output — Planning blueprint

Study area	Priority	Suggested weekly focus	Planning rule
Java programming	High	6 hours	Practise coding regularly; include OOP and problem-solving.
DSA	High	7 hours	Combine concept review with coding problems and timed practice.
SQL / DBMS	High	5 hours	Alternate SQL queries with DBMS concepts and short tests.
AI/ML fundamentals	Medium	4 hours	Use concept → example → short recall; connect concepts to applications.
Project work	Medium	4 hours	Use protected blocks for implementation, testing, and notes.
Daily revision	Essential	3 hours across the week	Use active recall and error review instead of only rereading.

Planning rules carried into Step 2

1. Schedule demanding coding or problem-solving work during the first available focus blocks.
2. Use approximately 50 minutes of focus followed by a 10-minute break.
3. Include a short active-recall session every day.
4. Pair a concept session with a related practice session when possible.
5. Spread project work across multiple manageable blocks rather than postponing it to one long session.
6. Use Saturday and Sunday for timed practice, weak-area correction, and weekly consolidation.

Step 2 output — First-draft schedule

Day	Draft focus blocks	Revision / outcome
Monday	Java coding; DSA concepts; SQL queries; AI/ML basics	Recall key syntax and solve one DSA problem again.
Tuesday	DSA problem-solving; Java OOP; DBMS; project implementation	Review mistakes and record project progress.
Wednesday	SQL practice; Java coding; AI/ML examples; DSA practice	Mixed recall quiz covering the day's concepts.
Thursday	DSA problems; SQL/DBMS; Java interview questions; project work	Revise DBMS definitions and Java interview answers.
Friday	Java mock coding; SQL test; AI/ML review; timed DSA practice	List weak areas to address over the weekend.
Saturday	Deep DSA; Java problem set; SQL/DBMS test; AI/ML application; project; weak-area repair	Weekly consolidation and project buffer.
Sunday	Coding mock; SQL/DBMS mock; AI/ML review; project completion; DSA revision; next-week planning	Summarize progress and set next week's priorities.

4. Critique Gate & Final Refined Output

Step 3 self-critique

Checklist item	Review finding	Correction applied
Hour limits	The first draft lists activities but does not consistently show how each block adds up to the requested focused hours.	Use four 50-minute focus blocks on weekdays and six on each weekend day, with breaks outside focused-study time.
Coverage of priorities	All five areas appear, but the amount of time assigned to each is not always explicit.	Assign each focus block a clear subject and retain regular project sessions.
Daily revision	Revision is mentioned, but the exact daily placement is vague.	Reserve the final 20–30 minutes of the day's study period for active recall and error review.
Session duration	The draft does not define a consistent focus-and-break cycle.	Use 50-minute focus blocks and 10-minute breaks; avoid treating breaks as focused hours.
Project buffer	Project work is present but could be pushed aside when technical practice runs long.	Protect two weekday project blocks and one longer weekend block.
Learning sequence	Some topics are listed without a clear connection between learning and application.	Follow concept review with related problems, queries, code, or a small project task.

Final corrected seven-day plan

The schedule below uses the requested number of focus sessions each day. Breaks are taken between sessions and are not counted as focused study time. Use the final review period within the day's available study allocation.

Day / target	Focused study sessions	End-of-day review
Monday 4 hours	1. Java coding and OOP 2. DSA concepts + two problems 3. SQL query practice 4. AI/ML fundamentals	20 minutes of active recall: Java syntax, SQL queries, and one DSA mistake.
Tuesday 4 hours	1. DSA problem-solving 2. Java collections / OOP practice 3. DBMS concepts 4. Project implementation	20 minutes: recall DSA patterns and summarise project progress.
Wednesday 4 hours	1. SQL practice 2. Java coding 3. AI/ML concepts with examples 4. DSA timed practice	20 minutes: mixed recall quiz and error log update.
Thursday 4 hours	1. DSA problems 2. SQL + DBMS review 3. Java interview questions 4. Project implementation/testing	20 minutes: explain DBMS concepts and Java answers without notes.
Friday 4 hours	1. Java mock coding 2. SQL mini-test 3. AI/ML revision 4. DSA timed practice	20 minutes: list weak topics and select weekend priorities.
Saturday 6 hours	1. Deep DSA practice 2. Java problem set 3. SQL/DBMS test 4. AI/ML application 5. Project work 6. Weak-area repair	30 minutes: review errors and consolidate the week's learning.
Sunday 6 hours	1. Coding mock test 2. SQL/DBMS mock test 3. AI/ML review 4. Project completion / testing 5. DSA revision 6. Weekly recap and next-week planning	30 minutes: summarise achievements, unfinished work, and next week's goals.

5. Dependency Verification & Evaluation

The following table checks whether the workflow satisfies the intended multi-step structure.

Requirement	Evidence in this assignment
At least three prompts	Three separate prompts are included: planning, generation, and critique/refinement.
Explicit dependency	Prompt 2 uses the Step 1 blueprint; Prompt 3 reviews and corrects the Step 2 timetable.
Critique gate	The third prompt requires a checklist-based review before the final timetable is produced.
Intermediate outputs	A planning blueprint and a first-draft timetable are shown before the refined schedule.
Practicality	The final schedule identifies study areas, uses consistent focus sessions, and includes daily revision.
Live evidence	Evidence must be captured from the actual AI chat after running the prompts in sequence. No screenshot should be fabricated or represented as live evidence if it was not captured from the interface.

Evaluation summary

The three-stage workflow demonstrates how a broad requirement can be refined through planning, schedule generation, and critical review. The review step improves the draft by making hour limits, focus-block structure, project time, and daily revision more explicit. In an actual execution, the AI's intermediate responses should be retained so the dependency between prompts can be verified.

Live evidence checklist

Before submitting, capture genuine screenshots of the AI interface showing the sequential interaction. The evidence should show: (1) Prompt 1 and its response, (2) Prompt 2 and its response using the planning blueprint, and (3) Prompt 3, the critique, and the corrected timetable. Place the screenshots in the evidence appendix or attach a verifiable chat link if your instructor permits it.

Evidence item	What to capture	Status
Prompt 1	Prompt and planning blueprint response	To be captured
Prompt 2	Prompt and first-draft timetable response	To be captured
Prompt 3	Critique checklist and final refined timetable	To be captured

Conclusion

This assignment presents a complete planning-to-refinement structure for an AI Study Planner. The key outcome is not merely a timetable, but a documented method in which each stage builds on earlier work and the final answer is checked against clear constraints. Actual live-interface screenshots should be added after executing the prompt chain.

Submission readiness: Student details and assignment content are prepared. Add authentic live AI-chat evidence if it is required by the evaluator.